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## Integrable Systems — Exercise Sheet 2

Two languages: the Poisson bracket and differential forms

Variant: Questions only

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Conventions, as in Lecture 2:  $\{f, g\} = \sum_i (\partial_{q_i} f \partial_{p_i} g - \partial_{p_i} f \partial_{q_i} g)$ , so  $df/dt = \{f, H\}$ ;  $X_f h = \{h, f\}$ , so  $X_f = \sum_i (\partial_{p_i} f \partial_{q_i} - \partial_{q_i} f \partial_{p_i})$ ;  $[X, Y]^i = X(Y^i) - Y(X^i)$ ; and  $(\alpha \wedge \beta)(v, w) = \alpha(v)\beta(w) - \alpha(w)\beta(v)$ .

**Exercise 1 (brackets and angular momentum).** On  $T^*\mathbb{R}^3$  let  $L = q \times p$ , that is  $L_1 = q_2 p_3 - q_3 p_2$  and cyclically. (i) Compute  $\{q_i, p_j\}$ ,  $\{q_i, q_j\}$  and  $\{p_i, p_j\}$ . (ii) Show that  $\{L_i, L_j\} = \sum_k \epsilon_{ijk} L_k$ ; in particular  $\{L_1, L_2\} = L_3$ . (iii) Show that each  $L_i$  Poisson-commutes with  $|q|^2$ ,  $|p|^2$ ,  $q \cdot p$  and  $|L|^2$ . Compute  $\{L_3, q_1\}$  and  $\{L_3, q_2\}$ , and say what the flow of  $L_3$  does. (iv) Deduce that for every central force,  $H = \frac{1}{2}|p|^2 + V(|q|)$  with  $V$  smooth, on  $q \neq 0$ , each  $L_i$  is a first integral, and state the Noether reading of this.

**Exercise 2 (the engine, in coordinates, for one pair).** On  $T^*\mathbb{R}$  let  $f = q^2 p$  and  $g = qp^2$ . (i) Compute  $X_f$  and  $X_g$ . (ii) Compute  $[X_f, X_g]$  from the component formula. (iii) Compute  $\{f, g\}$  and  $X_{\{f, g\}}$ , and check the engine  $X_{\{f, g\}} = -[X_f, X_g]$ . Which sign would the convention  $X_f h = \{f, h\}$  produce?

**Exercise 3 (first integrals, and Noether).** On  $T^*\mathbb{R}^2$  take the anisotropic oscillator  $H = \frac{1}{2}(p_1^2 + p_2^2) + \frac{1}{2}(q_1^2 + 4q_2^2)$ , with frequencies 1 and 2, and let  $H_1 = \frac{1}{2}(p_1^2 + q_1^2)$ ,  $H_2 = \frac{1}{2}(p_2^2 + 4q_2^2)$ ,  $L = q_1 p_2 - q_2 p_1$ . (i) Which of  $H_1$ ,  $H_2$ ,  $L$  are first integrals? Do  $H_1$  and  $H_2$  Poisson-commute? For the isotropic oscillator  $\frac{1}{2}(p_1^2 + p_2^2 + q_1^2 + q_2^2)$ , is  $L$  a first integral? (ii) Find the flows of  $H_1$  and of  $L$ , and check the Noether reading:  $H$  is invariant under the one and not under the other. Compute  $\frac{d}{ds} H$  at  $s = 0$  along the flow of  $L$ .

**Exercise 4 (covectors).** (i) On  $\mathbb{R}^2$  let  $f = x^2 + y^2$ . Write  $df$  in the basis  $dx, dy$ , and evaluate  $df$  at the point  $(1, 2)$  on the vectors  $v = (3, 1)$  and  $w = (-2, 1)$ . Interpret the zero in the lecture's picture of a covector as a stack. (ii) In polar coordinates  $x = r \cos \theta$ ,  $y = r \sin \theta$ , express  $dx$  and  $dy$  through  $dr$  and  $d\theta$ , and deduce that  $x dy - y dx = r^2 d\theta$  and  $x dx + y dy = r dr$ . (iii) Show that the 1-form  $p dq$  on  $T^*\mathbb{R}$  is not  $dF$  for any smooth function  $F(q, p)$ .

**Exercise 5 (the wedge).** (i) Evaluate  $(dx \wedge dy)(v, w)$  for  $v = (1, 2)$ ,  $w = (3, -1)$ , and  $(dx \wedge dy)(w, v)$ . Which way round do  $v, w$  run? (ii) On  $\mathbb{R}^2$ , compute  $(x dx + y dy) \wedge (-y dx + x dy)$ . (iii) On  $\mathbb{R}^3$ , write  $(dx + dy) \wedge (dy + dz)$  in the basis  $dx \wedge dy$ ,  $dx \wedge dz$ ,  $dy \wedge dz$ . (iv) On  $\mathbb{R}^4$  with  $\omega = dq_1 \wedge dp_1 + dq_2 \wedge dp_2$ , show that  $\omega \wedge \omega = 2 dq_1 \wedge dp_1 \wedge dq_2 \wedge dp_2$ , and that adding  $c dq_1 \wedge dq_2$  to  $\omega$  ( $c$  constant) does not change it. (v) Show that  $\alpha \wedge \alpha = 0$  for every 1-form  $\alpha$ , and give a 2-form  $\beta$  with  $\beta \wedge \beta \neq 0$ .

**Exercise 6 ★. The Jacobi identity with variable coefficients.** On  $\mathbb{R}^3$  with coordinates  $z = (x, y, z)$ , let  $J^{\alpha\beta}(z) = -J^{\beta\alpha}(z)$  be smooth, and define  $\{u, v\} = \sum_{\alpha, \beta} \partial_{\alpha} u J^{\alpha\beta} \partial_{\beta} v$ , so that  $J^{\alpha\beta} = \{z^{\alpha}, z^{\beta}\}$ . The lecture's reading proves the Jacobi identity when  $J$  is constant. (i) Show that for variable  $J$  the terms with second derivatives still cancel, and that the cyclic sum  $\mathcal{J}(f, g, h) = \{f, \{g, h\}\} + \text{cyclic}$  satisfies  $\mathcal{J}(f, g, h) = \mathcal{J}(x, y, z) \det [\partial_{\alpha} f, \partial_{\alpha} g, \partial_{\alpha} h]$ . So Jacobi holds for all functions if and only if it holds for the three coordinates. (ii) Take  $\{x_i, x_j\} = \sum_k \epsilon_{ijk} x_k$ , the bracket of Exercise 1 (ii) with the  $L_i$  as coordinates. Show that it satisfies Jacobi, and that  $|x|^2$  Poisson-commutes with every function. (iii) Show that  $\{x, y\} = 1$ ,  $\{y, z\} = 0$ ,  $\{z, x\} = x$  violates Jacobi.